Harsh Pathak

cs25mtech11010@iith.ac.in | harsh.pathak.hp1@gmail.com | +91 7291832396 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.45
Mount Carmel School, Dwarka, New Delhi	Class XII (CBSE)	2016	94.60%
Modern Convent School, Dwarka, New Delhi	Class X (CBSE)	2014	10.00%

PROFESSIONAL EXPERIENCE

Sabbatical: Transition to Graduate Studies

April 2024 - June 2025

Left industry for full-time preparation for GATE (CSE and DA) and TIFR Integrated PhD; qualified in both and joined the M.Tech program at IIT Hyderabad.

Goldman Sachs

Engineering Associate, July 2020 - March 2024

Software Engineer within Identity and Access Management Division.

- Core engineer on OCES, a cloud-native and policy-driven centralized authorization platform for the firm, featured on the Goldman Sachs Developer blog
- Engineered data pipelines to allow policy and reference data ingestion from on-prem systems to cloud
- Provisioned and managed cloud infrastructure using Terraform IaC
- Designed and developed core features such as Bundle Discovery and Bundle Consolidation, reducing client onboarding effort.
- Implemented end-to-end observability using Datadog and Prometheus, instrumenting application metrics, centralizing log ingestion, and building dashboards and alerts to monitor microservices health and performance.
- Automated CI/CD pipelines that enabled policy authors to test, validate, and deploy policies to the cloud through a self-service workflow, eliminating manual release steps.
- Reduced monthly AWS and Datadog costs by 40%, rightsizing compute resources using utilization metrics and reducing log volume at the source.
- Reported 6 issues to Open Policy Agent while using it in production for OCES. All issues were acknowledged and addressed by maintainers.
- Authored design documents, user guides, and technical documentation, supporting architecture decisions and user onboarding.

Adobe

Software Engineering Intern, May 2019 - July 2019

- Integrated Progressive Calibration Networks on Adobe Lightroom to detect rotated faces in photos.
- Compared PCN's performance with existing in-house solutions on multiple face detection datasets like LFW, WIDER Face, etc.
- PCN delivered better detection accuracy with faster inference and a lower memory footprint.

SKILLS

Tools and Technologies: Apache Kafka, Terraform (IaC), AWS, CI/CD, Open Policy Agent, Datadog, Prometheus, Redis, Docker

ML/DL Frameworks: PyTorch, Scikit-learn, NumPy

Programming Languages: Java, Python, C++

SCHOLASTIC ACHIEVEMENTS

- 2025 Qualified GATE CSE and DA (Data Science and AI) exams

- 2025 Qualified the written exam for TIFR Integrated PhD program in Theoretical CS

- 2019 Secured Dean's List Status (A.Y. 2018-2019, at IIIT Delhi) for excellence in academic performance

- 2016 Qualified JEE Mains and JEE Advanced

RESEARCH EXPERIENCE

M.Tech Thesis

Advisor - Dr. J. Saketha Nath and Dr. P. N. Karthik, IIT Hyderabad

May 2026 - Present

Research in collaboration with DLRL Hyderabad on AI-driven threat analysis and automated countermeasure response.

B.Tech Project

Advisor - Dr. A. V. Subramanyam, IIIT Delhi

January 2019 - June 2020

Researched and developed Deep Learning models to detect Deep-fake videos [Report]

TEACHING EXPERIENCE

Teaching Assistant, IIT Hyderabad (M.Tech TA-ship)

- **ID2230 Data Structures & Applications**, Jul - Nov 2025: Graded assignments and exams. Instructor(s) - Dr. M. V. Panduranga Rao and Dr. Rakesh Venkat
- **CS2030 Theory of Computation**, Jan - May 2026: Graded assignments and exams. Instructor - Dr. Nitin Saurabh
- **CS2233 Data Structures**, Jul - Nov 2026: Conducted lab sessions and tutorials, graded assignments and exams. Instructor - Dr. Manaswi Paraashar

PROJECTS

BookIT - Classroom Bookings Management System

August 2017 - December 2019

Guide - Dr. Vivek Kumar, IIIT Delhi

Jointly developed a classroom booking system for the CSE201 course (Aug - Dec 2017); extended through 2019 after faculty appreciation. Recommended for managing classroom reservations at IIIT Delhi. [Video] [Code]

ARM Decompiler Application

August 2017 - December 2017

Developed a decompiler for ARM in C++ that translates ARM code to C language. [Code]

Reinforcement Learning on Traffic Lights

August 2019 - December 2019

Implemented a deep Q Learning algorithm on PyTorch to minimize average waiting time for vehicles in multiple traffic simulations created using SUMO. [Code]

Deep-fake Creation

January 2019 - April 2019

Implemented multiple adversarial training algorithms to mimic facial expressions using PyTorch. [Presentation]

Hand Segmentation for Cursor Control

January 2019 - April 2019

Applied Optical Flow and Morphological Snakes-based Techniques to segment hands in webcam images and achieve cursor control. [Code]

RELEVANT COURSES

Graduate Level Courses (at IIT Hyderabad, 2025-2027)

Advanced Data Structures and Algorithms (CS6013), Foundations of Machine Learning (CS5590), Advanced Computer Networks (CS5060), Cryptology (CS6160), Network Security (CS6903), Deep Learning (CS5480), Reinforcement Learning (AI7010)

Undergraduate Level Courses (at IIIT Delhi, 2016-2020)

Data Structures and Algorithms (CSE102), Computer Organization (CSE112), Advanced Programming (CSE201), Fundamentals of Database Systems (CSE202), Analysis and Design of Algorithms (CSE222), Operating Systems (CSE231), Computer Networks (CSE232), Machine Learning (CSE343), Computer Vision (CSE344), Deep Learning (CSE641), Digital Image Processing (CSE340)

EXTRACURRICULARS

Community Service

September 2019 - January 2020

Volunteered with Choti Si Khushi, a non-profit dedicated to empowering underprivileged communities through education, healthcare, and livelihood support. I taught Math and Science courses to students (classes 4-9), helped conduct the Annual Sports Meet and designed their annual newsletter.